# LINEAR GRAPHS: PLOTTING $Y = MX + C$

Grade 5 Mathematics | Worksheet & Practice Exercises

**Name:** ______________________ **Class/Sec:** ____________ **Date:** ____________

## 1. Lesson Concept Recap

**Meet Rohan's Auto Ride!**

Rohan takes an auto-rickshaw to the cricket ground. The driver charges a fixed base fare of ₹23 just to start the meter, plus ₹10 for every kilometer traveled!

**Formula:** $\textit{Total Fare} = \textit{Base Fare} + (\textit{Rate} \times \textit{Distance}) \rightarrow y = 10x + 23$

**The Magic Equation:** $y = mx + c$

- $x$ = Input value (e.g., Distance)
- $y$ = Output answer (e.g., Total Fare)
- $m$ = Steepness / Gradient (rate of change / step size)
- $c$ = Starting Value / y-intercept (where line touches y-axis at $x = 0$)

**Coordinate Map Basics:**

- **x-axis:** Horizontal line (left to right)
- **y-axis:** Vertical line (bottom to top)
- **Origin (0,0):** Where axes cross
- **Point Address:** $(x, y)$

## 2. Guided Calculation Example: $y = 2x + 1$

Find $y$ by substituting values of $x$ into the equation to get coordinates $(x, y)$:

| Input ($x$) | Calculation ($2x + 1$) | Output ($y$) | Coordinate Point ($x, y$) |
|---|---|---|---|
| 0 | $2(0) + 1$ | 1 | (0, 1) |
| 1 | $2(1) + 1$ | 3 | (1, 3) |
| 2 | $2(2) + 1$ | 5 | (2, 5) |
| 3 | $2(3) + 1$ | 7 | (3, 7) |

## 3. Practice Questions

### Q1. Warm-Up Challenge: Predict the Y-Value

If $y = 3x + 2$, what is the value of $y$ when $x = 2$?

[ ] A) 6 [ ] B) 8 [ ] C) 10 [ ] D) 12

### Q2. Identify $m$ and $c$

For the equation $y = 4x + 7$, identify the gradient (steepness $m$) and the y-intercept (starting value $c$):

**Gradient ($m$):** ____________ **Starting Value ($c$):** ____________

### Q3. Complete the Table & Find Coordinates

Complete the calculation table below for the line $y = 2x + 3$:

| Input ($x$) | Calculation ($2x + 3$) | Output ($y$) | Point ($x, y$) |
|---|---|---|---|
| 0 | $2(0) + 3$ | _____ | (0, _____) |
| 1 | $2(1) + 3$ | _____ | (1, _____) |
| 2 | $2(2) + 3$ | _____ | (2, _____) |

### Q4. Real-World Application: Rohan's Return Trip

On his return trip home, Rohan takes a different taxi that charges a base fare of ₹15 and a rate of ₹8 per kilometer ($y = 8x + 15$).

a) What is the total fare ($y$) if Rohan travels $x = 5$ kilometers?

b) Write down the coordinate point $(x, y)$ for this 5 km trip: _____________

### Q5. True or False?

State whether the following statements are **True** or **False**:

a) The point (0, 0) is called the origin. [ ]

b) In $y = mx + c$, $c$ represents how fast $y$ grows for every step of $x$. [ ]

c) For the line $y = 5x + 2$, the line crosses the y-axis at $y = 2$. [ ]